



MODULE 1 UNIT 1

Activity submission scenario

UNIVERSITY OF CAPE TOWN
BUSINESS SYSTEMS ANALYSIS



Instructions

This scenario will run throughout the course and will form the basis of your activity submissions. The scenario will be expanded on throughout the course and you will be told explicitly in the activity submissions which sections of the case study the activity submission is based on. Quiz questions will not be based on this scenario unless stated otherwise.

Note:

This scenario is entirely fictitious. Although many details are inspired by real life, resemblance to any real-life person, entity or event is purely coincidental and unintentional.

You might find this scenario detailed with seemingly irrelevant information. However, as a business systems analyst (BSA), you will need to work with significantly more information than this, whether related to business and system needs, or interpersonal aspects and politics. The scenario in this document, therefore, mimics reality in that it requires you to discern which information is relevant in each situation, and which might be relevant later, or which is entirely irrelevant. As the scenario unfolds throughout the course, you may want to revisit some of the earlier scenario documents as the details they contain could become relevant later on.

In this and subsequent scenarios in this course, you will notice the use of “we” as well as “you”. “We” here refers to you and your client or employer, or what the business or solution must do, depending on the context.

Scenario

Today is the Friday that ends the first week of your contract as a new business systems analyst (BSA) at a new client. Your client is an online fishing products retailer named Kabeljou (after the Afrikaans name for Silver Kob, a coastal fish) and is pronounced *cub-bill-yoh*. You are the first BSA that Kabeljou has hired.

Your role in the company

You responded to an online job advertisement for a BSA. The ad was somewhat vague about the precise duties of the position, but what appealed to you most were the opportunities for a high degree of freedom in designing new or improved systems to replace manual and fatigued systems, and that these would be critical to growing a young, dynamic and successful online retailer.

You’re not sure who your line manager is because your interviews were conducted by multiple people. Although Miriam, the finance manager, signed your contract, you were told that she does this for all new hires. Most of your colleagues are keen to support you in settling in, but no-one, in particular, is assigning you tasks or expecting any progress updates from you.



During your recruiting interviews and your first week's efforts in meeting your colleagues and discovering Kabeljou's systems, you note that the business's computer systems are like spaghetti and short bread – a tangle of disparate systems, with crumbly code and processes. The need for new or replaced systems in many areas of the company is obvious. You could not have joined Kabeljou at a more opportune time to have your professional prowess be put to good use. Your first project will be the replacement of the current website with an e-commerce site.

Below are the highlights of what you've learned so far in your new job.

Company overview

Kabeljou was formed about three years ago by two colleagues – Marius Kruger, now the managing director (MD), and Anton Wolfaardt, the chief operations officer (COO). Establishing the company was tough: both the events leading up to the new venture, as well as the start-up period itself. Although they had a rocky start, they seem to have been successful for the past two years. It is difficult to pinpoint exactly when "tough" turned into "successful", but do not let the company's current success lull you into believing that times are easy. Things at Kabeljou are anything but easy, as you're discovering.

Of the roughly 20 staff members, most are full-time employees, with about a handful who are part-time and contract workers like you. The company recently moved into a new warehouse with an adjoining head office in Paarden Island, Cape Town. Half of the employees are co-located at the warehouse and head office, while the remainder are home-based employees from across South Africa, several from as far away as KwaZulu-Natal and Mpumalanga.

Marketing

From a marketing perspective, Kabeljou's niche (their product range and target market) is the retail supply of sea fishing and angling tackle (lines, rods, reels, baits and lures, sinkers, hooks, et cetera) for leisure, competition, and micro-commercial fishermen in South Africa. In reality, while most deliveries are within South Africa, about 12.5% of their few thousand monthly orders are foreign, mostly from across Southern Africa, but with a few orders placed weekly by loyal customers further away.

About two thirds of all orders placed by customers are delivered to them through a post office counter-to-counter service. The remainder are delivered by two courier service providers – one for express domestic and the other for all international destinations.

In the beginning, Marius and Anton built their customer base through networking directly with fishermen at the harbours and at events like boat shows and fishing competitions. This is how they tested their business idea before formally founding the company. Within a few weeks of founding, they started their first tactic for online lead generation. This was SEO-driven and attracted customers through organic search.



Now, for about the past year, they've been publishing a monthly email newsletter. This newsletter is a product catalogue supported by one or two promotional articles in each issue. The mailing list consists of 5,011 subscribers and is growing by about 100 people per month (mostly from sign-ups through the website). One of the improvements you think could be made to the newsletter is to make it more customer oriented, rather than product focused.

Opportunities and threats

General top-priority opportunities relate to the growing online customer base across all industries and the acceptance of online shopping as a way to shop. Improved online security and payment methods, including mobile payment systems, have contributed to this growth.

Possible market-related strategies include a market expansion strategy of targeting new segments, like bigger commercial fisheries in South Africa, or leisure and micro-commercial fishermen in other African countries, or even non-African customers, like fishing safari tourists. Market penetration strategies, on the other hand, could involve increasing the value of each order placed, or increasing the frequency of each customer's orders, or both.

The product-oriented strategic opportunities that Marius highlighted involve new lines in fishing apparel, fly fishing tackle and apparel (which could simultaneously open a new market segment of inland customers), and even venturing into online boat sales (although that option is already heavily competed).

Speaking of competition, threats to Kabeljou's future are most likely (in Marius's view) to come from new entrants to online selling – both specialist and chain retailers entering the niche.

An ongoing challenge has been the bungling of fishing licences and quotas by a government department. This has become highly politicised in recent years, probably because of the perception that large fishing companies are favoured over small fisheries, and the detrimental impact of this on the livelihood of traditional fishermen, threatening to turn some small fishing villages into ghost towns. Although this has no direct impact on Kabeljou, the effects are now felt strongly through the closure of some of the micro-fishing businesses, many of which had been regular and high-volume customers since Kabeljou's opening days. As much as Marius and Anton are in business for financial benefit, they'd like to ensure that their systems and products remain accessible to even the poorest of fishermen. What this intention means for Kabeljou and, for you as a BSA in particular, is that the company cannot afford elaborate, complicated and expensive systems, and that customer-facing systems must be as easy to use as possible.

You also face two other problems. The first is legislative compliance, especially the new Protection of Personal Information Act (POPI) that could severely hamper online marketing. One of the major areas of online marketing that POPI will affect is that spam will be outlawed and emailing a prospective customer is allowed only with the recipient's permission.



Secondly, internal growth of staff numbers at Kabeljou threatens to add either bureaucratic overhead or, if nothing is done about the current crisis, lowering of morale and further confusion caused by inadequate internal communication and ineffective business processes. This will likely hamper business performance through lower lead generation numbers and higher customer churn through poor service levels.

More about who's who

You've spent much of your time in your first week with Marius, Anton and the other three managers. You've heard many personal stories, and you've also discovered very useful business information.

Marius and Anton, the co-founders, had worked together previously and refer to each other as colleagues. They are, in fact, close family friends and have a history that goes back about 15 years.

The story is that **Marius**, who is now 32 years old and lives in upmarket Sunset Beach, enjoyed a privileged childhood. Despite being born into upper-middle class wealth, he chases his own goals and is a down-to-earth, approachable man. You've found his demeanour affable and relaxed, dressing like a surfer and sporting sunglasses like an Alice band. Though hard-working and far from ostentatious, Marius is not shy to treat himself to water sports toys and surfing holidays at exotic destinations. His competitive spirit (he once represented South Africa at an under-19 surf-ski tournament) is the energy behind Kabeljou.

Marius' father, Magnus, owned a fishing business with a fleet of six boats. During school and university holidays, Marius worked as a deckhand, fisherman, and eventually as a first mate on the boats, almost always sailing with Anton. Anton, who is now in his mid-fifties, enjoyed working with Marius, and is like a second dad to him.

Tragedy struck five years ago when Magnus died of heart failure. That's when Marius took over running the family business, relying heavily on Anton's leadership, wisdom, and their strong relationship for mutual support.

That was not the only disaster. Four years ago, they lost 11 crew members and two boats in a storm, and three other boats were severely damaged. Unable to recover from the disaster and Magnus' death, they sold the remaining assets and shut down the business. The ensuing depression was tough, but, with their fishing expertise and network of contacts, they found new energy in launching Kabeljou – a "vessel" with much less physical risk.

Marius oversees all marketing functions at Kabeljou and, as head of marketing, initiated the project to develop the e-commerce system. He prefers to work from home and, although he pops in at the head office at least every second day, he has no dedicated office of his own. In response to your asking what his biggest complaint is, he lamented that they (the executives) receive information too slowly. What might be his magic wand solution? He wants reports such as cash flow forecasts, customer and orders statistics, warehouse stock levels, dispatch



performance, and routine monthly financial reports much sooner after their underlying events than he is currently receiving them.

Strategically, Marius wonders if they should explore vertical integration more seriously. He is looking upstream, like buying out a manufacturer, or buying from a white-label manufacturer and launching a Kabeljou-branded range of fishing equipment, or even starting their own factory. Downstream in the value chain, he has explored running their own courier operations.

As a gadget geek, Marius has even suggested developing products that can be emailed and 3D-printed in order to cut distribution costs. Of course, not only is the market for this still tiny, but these cost savers will be available to their competitors just as easily, so this doesn't offer a strategically competitive advantage. More relevant to reality today, though, Marius has decreed that the new e-commerce system must be mobile-friendly.

Anton's back story and Marius' are like chalk and cheese; they have little in common. Having grown up in a fishing village near Veldrif on the cold-water western coast of South Africa, Anton's childhood was simple and austere, typical of local West Coast communities of indigenous descent. His relationship with sea fishing is almost as old as he is, which shows on his weathered and determined features. His father owned and fished from a shore-launched dory and he literally showed Anton the ropes. He used these skills to skipper for Magnus and quickly advanced to fleet and operations manager with Magnus.

Anton prefers the old-fashioned approach to management by walking about the office and checking on employees. He is invariably first in and last out of the office. He admits that he struggles to connect with the employees and freelance consultants who work remotely from home, which you suspect is part of what's behind the subtle complaints of bad communication at Kabeljou. Anton runs a tight ship, so to speak, but has become comfortable with a relaxed culture of allowing people to prove themselves within boundaries. The boundaries at Kabeljou are very dynamic because of the company's growth, and therefore, not always clear.

Being a rapidly growing business, Marius and Anton have employed many people within a short period of time. Here are some of the managers who are taking over some of the founders' responsibilities:

Kadys is the warehouse manager. He grew up on the Cape Flats. He is in his late twenties and has a diploma in Operations Management. Previously, he was a store supervisor for a retail chain selling outdoor leisure and camping equipment. A key priority for him currently is the purchase of the new pallet trucks to replace the old ones that came with the warehouse because the old trucks are in continual need of repairs. He is also pestering Vuyelwa and Miriam to implement a barcode tracking system and barcode readers for the warehouse pickers and packers.

Vuyelwa is the information technology (IT) manager and, like Kadys, is in her late twenties. She gained some junior management experience after she moved from Kwa-Zulu Natal to complete her BSc in Computer Science at the University of Cape Town. She oversees all technical operational aspects of the current website and newsletter administration, and she



manages all IT-related projects at Kabeljou. One of the problems that she shared is the resistance that some managers have to her suggestions to “upgrade” from their spreadsheets and paper systems to integrated, web-based services. She sees you as an ally to her cause and, because of this, she has confided in you about a few other, seemingly irrelevant, personal details.

Although about half of the remote employees do IT-related work, like catalogue maintenance, the virtual call centre, website updates and newsletter production, and although they work closely with Vuyelwa, only the three customer service representatives (CSRs) in the virtual call centre report to her directly. In fact, it’s not clear who their line manager might be. You suspect that there is probably little or no performance management for employees.

As the finance manager, **Miriam** handles almost everything related to Kabeljou’s finances, importing and customs clearances, supplier contracts, payments, as well as billing and collections. Marius and Anton only get directly involved in finances for the very big transactions and at their executive committee (“exco”) management meetings, which are supposed to be held every Friday.

Miriam and Kadys were neighbours when Kadys grew up and he still addresses her as “aunty”, which has rubbed off onto some of their colleagues. She introduced Kadys to Kabeljou. She has a tight rein over the purse strings, but some of her opinions and practices tell you that she might be too risk averse for a company that needs a lot more risk tolerance and a more contemporary approach to technology to support its growth as an e-tailer. For example, not all the financial data Miriam shared with you was printed – some financial summaries were hand-written on loose sheets of notepad pages. You realised that you’d overestimated Kabeljou’s computer system maturity when she remarked that she prepares these hand-written “reports” for each exco meeting.

Because there’s no human resources (HR) manager, Miriam also handles payroll and related HR admin, such as leave tracking and employment contract filing. This is yet another example of Kabeljou’s system immaturity relative to their organisational size and requirements. All salary calculations, tax returns and pay records are done in Excel spreadsheets and stored on Miriam’s laptop. The monthly tax returns are done online and not on paper, however. You ponder that surprisingly, for an online retailer, there’s so much that is done manually.

All five managers can be considered super users of Kabeljou’s website, catalogue and billing computer systems. Although key parts of these systems, like the debtors reconciliation spreadsheets, are so haphazard and unsystematic, they can be called “computer systems” only by definition. This role of super user can be expected to extend to the new e-commerce system – you will need to work with all five of them closely during this project.

New strategies needed

Both Marius and Anton see the new e-commerce platform as a key tactic to whatever new business strategy they might devise, regardless of whether the strategy will be market led or product oriented.



They are open to affiliate marketing as an approach to capitalise on their personal network. They also want to build a media-buying function into their marketing activities, which involves techniques such as making and placing banner ads, paid search, pay-per-click ads, and so-called solo ads (buying ad space or advertorials in the blogs and emails of popular bloggers). The newsletter is also not immune to being overhauled. They want to introduce edutaining articles and videos that are unusual and remarkable and hence, potentially viral, as a way to both increase traffic and retain the interest of existing customers.

They might also want to run campaigns for existing customers only, and they are considering a customer relationship management (CRM) system to help reduce customer churn. This, they believe, needs to focus on nurturing customer relationships and thereby increase customer lifetime value (CLV), rather than quick sales that are also quickly lost to price-sensitive shoppers.

However, they aren't confident about whether the strategic priority is to increase traffic to keep filling the sales pipeline, to focus on lead conversion, or to drive customer retention in order to maximise CLV.

Kabeljou's products are well known in the market, so they are relatively easy to sell in early contact. In other words, compared with a product or service that is intangible or hard to explain, it is easy to convert a lead into a buying customer within, say, the first or second advert or newsletter they get. With a commodity-type product, though, it's easy to lose a long-standing customer to cheaper competition.

Marius concludes that deeper market research is needed, with a better intelligence of customer needs. He is keen to discover exactly what customers would be ready to pay a premium for.

Along the theme of having many fishing lines in many spots is common at Kabeljou, so to speak, they are also considering sponsoring events or individuals in fishing competitions. The negative factor in this brand-building approach is that costs can easily become exorbitant for a small business's budget. In addition, Anton is wary of the impossible task of accurately tracking offline "eyes on the brand" and how this converts to sales.

Going inside

From the customer's perspective, everything on Kabeljou's outside looks impressive, from the eye-catching logo to a wide range of specialist product images and write-ups in the well-presented online catalogue.

Looking inward, some internal problems are becoming more difficult to ignore. Marius and Anton are feeling increasingly disconnected from the sea-level staff and their customer base. They complain bitterly about becoming stereotypical executives who have no life outside of work.



They acknowledge that their internal structures and business systems are probably not keeping pace with their growth, but they're not sure how to remedy this or where to start. They admit that they would rather be "at sea level" with their customers than worrying about office, IT and warehouse issues.

Formalised business systems are in desperate need. The last, and only, semblance of a business plan is the two-page outline and handful of financial forecasts that Anton and Marius cobbled together more than three years ago when they founded Kabeljou. Policies and procedure manuals are inadequate. Job roles are vague and reporting lines are blurry. Although the better-organised staff make use of software tools and templates, these are usually created on the fly by each person and neither shared widely nor used consistently by others doing similar tasks. A less obvious risk, which only Vuyelwa alluded to, is that there are no business continuity systems, like remote data storage and disaster recovery procedures.

Operationally, there is a lot of miscommunication that causes re-work, double work, and no work. Many orders are delivered later than promised or not at all. Vuyelwa complains that, because customers can pay via an electronic funds transfer (EFT) and direct deposit, many payments can't be reconciled to their orders. This adds a huge burden on her CSRs who are overworked by the high volume of complaints for late product deliveries, which are then couriered door-to-door at the company's expense.

To add to management's woes, many low-level decisions wait on Marius or Anton for approval. Although this serves them by forcing information to flow to Marius and Anton, this not only makes for slow and expensive decisions (Anton's and Marius' time costs more than everyone else's time), middle managers are disempowered and decision quality is compromised, because Marius and Anton are no longer the experts in every situation.

Further internal challenges in the proverbial offing are likely to come from the urgent and clear need to grow the sales and marketing team. If market penetration or market expansion strategies are pursued, then many more people will be needed to create content for newsletters, handle media buying (for example, the placing of banner ads on web sites and sponsored links in social media), manage affiliates, and for a new type of CSR who needs proactive account manager skills and not reactive call centre, problem-solving skills.

If product expansion or diversification are strategies, Kabeljou will also need more people to handle supplier sourcing, product buying and product manager-type roles that include market research and getting out there with the end consumer. One thing is certain: Kabeljou is enjoying the second general business problem: that of having too many customers! (The first general business problem is too few customers.)

Priority requirements

To continue Kabeljou's growth and success, it is vital that the new virtual store provide end-to-end functionality. This means that, at a high level, features must span the life cycle of a typical transaction, being from sales landing pages to beyond the emailed order confirmation and everything in between. It needs an integrated product catalogue and shopping cart,



check-out and online payment, and self-service CRM through customer accounts with secure logins. Beyond the sale, customer service must extend to after-sales service and close the marketing loop with marketing analytics, customer feedback, and design of re-marketing campaigns for customer retention. With e-commerce, payment for each order would be directly on the system, which would eliminate our payment reconciliation function. This will improve service by reducing delivery times.

Based on conversations Marius and Anton had with a supplier about year ago, which led to Kabeljou getting a quote for an e-commerce system, Marius expects to budget around R250,000 for the e-commerce project. He thinks we can get an operational system within three months.

You and Vuyelwa believe this is somewhat unrealistic and you both agree that we should not be surprised if the budget is closer to R350,000, with a project duration of between five and eight months.

You have your work cut out for you – good luck!